the analogy that built chemical engineering

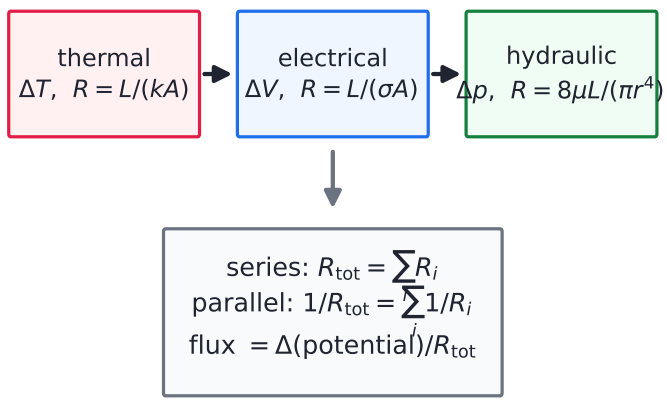
Reynolds · Prandtl · Chilton–Colburn · Bird–Stewart–Lightfoot

heat — $q = -k\nabla T$ — potential: T — flux ratio: Nu — Fourier

mass — $J = -D\nabla c$ — potential: c — flux ratio: Sh — Fick

momentum — $\tau = \mu\nabla u$ — potential: u — flux ratio: C_f — Newton

$$\text{flux} = -(\text{transport property}) \times \nabla(\text{potential})$$

circuit thinking

every *linear* steady transport problem is a resistor network — which is why finite-volume codes and bond graphs look the same